

Tutorial 11

April 15, 2026

Question 1

Explain how to solve the following problem:

$$\begin{aligned} \min \quad & (1/2)x^T x + c^T x \\ \text{s.t.} \quad & Ax \leq b \end{aligned}$$

using barrier method and Newton method. where $x \in R^n$, and $A \in R^{n \times n}, c, b \in R^n$ are given.

Question 2

Write down the Newton Equation in the barrier method for

$$\begin{aligned} \min \quad & (1/2)(x - a)^T P^{-1}(x - a) \\ \text{s.t.} \quad & \mathbf{0} \leq x \leq \mathbf{1} \end{aligned}$$

where $x \in R^n$, and $P^{-1} \in S_{++}^n, a \in R^n$ are given.